Transport in a product of type families agrees componentwise with the transports

The Univalent Foundations Program

hott-book:basics.tex:local-snapshot-20261014; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author theorem labelled thm:transprod (source lines 1356-1363); the proof span 1364-1373 is the author's proof environment, retained line for line.

Editorial note. The proof closes by invoking the product uniqueness principle derived from the product path characterization; both the characterization's statement and the author's uniqueness remark are retained as context.

Editorial note. The remark that the uniqueness principle is a special case is the author's own one-line derivation, retained verbatim in context.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: componentwise agreement of transport with the product family decomposition. Only the counted theorem and its author proof are retained.

Allowed standard prerequisites (editorial).

- Dependent type families over a base type and pointwise products of such families.
- Transport along a path, with its definitional behaviour on reflexivity.
- Path induction with its conversion rule.
- The path characterization of product types and its uniqueness corollary (the retained contexts).

Source reference. Original equation 2.6.3: sources/hott-basics/pinned-original-source.tex, source lines 1262–1262. The author's equation naming the inverse map, cited by the retained uniqueness remark. Navigation pointer only: the remark's content is retained verbatim and no part of the counted proof is replaced by the reference.

For any elements $x, y : A \times B$ and a path $p : x =_{A\times B} y$, by functoriality we can extract paths $\mathsf{pr}_1(p) : \mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)$ and $\mathsf{pr}_2(p) : \mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)$. Thus, we have a function

$$(x =_{A\times B} y) \to (\mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)). \tag{0.0.1}$$

Theorem 0.0.2. *For any x and y, the function* (0.0.1) *is an equivalence.*

In particular, we have shown that (0.0.1) has an inverse (2.6.3), which we may denote by

$$\mathsf{pair}^{=} : (\mathsf{pr}_1(x) = \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) = \mathsf{pr}_2(y)) \to (x = y).$$

Note that a special case of this yields the propositional uniqueness principle for products: $z = (\mathsf{pr}_1(z), \mathsf{pr}_2(z))$.

Theorem 0.0.3. *In the above situation, we have*

$$\mathsf{transport}^{A\times B}(p, x) =_{A(w)\times B(w)} (\mathsf{transport}^{A}(p, \mathsf{pr}_1 x), \mathsf{transport}^{B}(p, \mathsf{pr}_2 x)).$$

Proof. By path induction, we may assume p is reflexivity, in which case we have

$$\begin{aligned}\mathsf{transport}^{A\times B}(p, x) &\equiv x\\ \mathsf{transport}^{A}(p, \mathsf{pr}_1 x) &\equiv \mathsf{pr}_1 x\\ \mathsf{transport}^{B}(p, \mathsf{pr}_2 x) &\equiv \mathsf{pr}_2 x.\end{aligned}$$

Thus, it remains to show $x = (\mathsf{pr}_1 x, \mathsf{pr}_2 x)$. But this is the propositional uniqueness principle for product types, which, as we remarked above, follows from Theorem 0.0.2. □